Coral Bleaching Classification and Prediction

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Abstract

Coral reef bleaching is a growing environmental crisis driven largely by rising ocean temperatures, threatening marine biodiversity and coastal communities worldwide. Our project aims to support early detection and prediction of coral bleaching events using machine learning techniques to enable effective reef monitoring and inform conservation strategies. In this project, we develop machine learning models using two complementary datasets: an image dataset for visual classification of coral health and a structured environmental dataset for predicting bleaching occurrence and severity. For the image dataset, we implement a baseline binary logistic regression model to classify coral images, followed by a convolutional neural network, ultimately refined through architecture expansion and hyperparameter tuning. For the structured dataset, we use a multiclass logistic regression baseline model to predict bleaching occurrence based on environmental and geographic features, with additional models including a feed-forward neural network and histogram gradient boosting classifier. Model performance is evaluated using appropriate classification and regression metrics. Our results show that the refined convolutional neural network performs best on image data (75.14% test accuracy) while the histogram gradient boosting model yields the strongest results for our environmental dataset (79.75% test accuracy, 75.68% F1 score).

Introduction

Coral reefs are critical to both oceanic ecosystems and human beings, supporting nearly 25% of all marine species, while providing essential economic and environmental benefits, including coastal protection and food security [source]. Coral reefs also mitigate the impacts of storms and coastal erosion and play an important role in filtration of oceanic pollutants [source]. However, rising ocean temperatures and environmental stressors have led to an increase in coral bleaching events, a process in which corals expel the symbiotic algae that provide their color and primary energy source, which can result in mortality if conditions persist. The increasing frequency and severity of bleaching events pose a significant threat to marine biodiversity and the livelihoods of millions of people worldwide.

Given the scale and urgency of this problem, early detection and prediction of coral bleaching are critical for enabling effective conservation strategies. Machine learning offers a powerful approach to identifying complex patterns in both visual and environmental data on bleaching events. We investigate two key tasks: binary classification of coral health from images and predicting bleaching classes using environmental and geographic features. The input to our first model is an image dataset of coral reefs, and we use a baseline logistic regression model implemented in TensorFlow to predict a binary output indicating whether a coral is bleached or healthy. The input to our second model is a structured dataset consisting of numerical features such as reef coordinates, distance to shore, depth, sea surface temperature anomalies (SSTA), thermal stress anomalies (TSA), turbidity, and temporal indicators, along with categorical features including exposure and substrate type. We use logistic regression as a baseline to predict three bleaching severity classes: low (<1%), medium (1-30%), and high (>30%) - following the percentage class splits used by Chueng et al. (2025).

Related work

Coral bleaching prediction has traditionally relied on temperature-based metrics, with degree heating weeks (DHW) widely used to estimate heat-stress exposure. Cheung et al. (2025) adopted an interpretable ML approach (ordinal random forest + SHAP) to predict coral bleaching severity using 25 environmental predictors including thermal history, cloud cover, turbidity, and water flow to predict coral bleaching severity on the Great Barrier Reef. Their strength is capturing non-linear interactions and using spatial cross-validation to avoid overfitting (80% accuracy). A key insight from their work is that "reefs with weak currents or high water clarity showed higher bleaching risk even with moderate heat stress," highlighting the limits of temperature-only models. However, the model is region-specific and lacks fine-grained visual data.

A complementary line of research uses image-based deep learning to classify coral health. Earlier studies demonstrate that CNNs can detect bleaching from underwater imagery with high recall, though these models often struggle with domain shift and lack environmental context. Macrohon and Hung (2025) extend this line of work by benchmarking ResNet, Vision Transformers, and CNN architectures on a globally diverse underwater dataset, finding that a tuned CNN achieves state-of-the-art accuracy (88%) for bleaching classification. While these approaches demonstrate strong performance in visual classification, they remain primarily focused on pixel-level predictions without incorporating environmental drivers.

State-of-the-art approaches are increasingly moving toward multi-modal frameworks that integrate imagery with environmental time-series data, alongside explainable AI techniques to identify spatial patterns and climate refugia; our project aims to move closer toward this strategy. Our environmental feature set extends beyond temperature-based indicators to include spatial (latitude, longitude, depth, distance to shore), thermal (SSTA, TSA, DHW-related metrics), and environmental variables (turbidity, windspeed, cyclone frequency), enabling a more holistic representation of bleaching drivers. Unlike prior machine learning research on coral bleaching, our design assesses ML performance on both visual and structured datasets, positioning it closer to emerging multi-modal methods.

Dataset

This project uses two datasets: (1) a structured environmental dataset, and (2) an image dataset for coral classification. The BCO-DMO Global Bleaching & Environmental Dataset contains 41,361 observations collected between 1980 - 2020, with 62 environmental and geographic features describing coral reef conditions compiled from seven sources: Reef Check, Donner et al. (2017), McClanahan et al. (2019), FRRP, Safaie et al. (2018), and Kumagai et al. (2018). It provides broad spatial and temporal coverage of bleaching events and associated stressors such as temperature anomalies, cyclone frequency, turbidity, and exposure. The dataset, although initially cleaned by BCO-DMO, required additional preprocessing before modeling. Columns with few missing values were imputed, and a few duplicate observations were removed. Because the goal was to predict bleaching severity, we transformed the zero-inflated and right-skewed *percent_bleaching* variable into an ordinal class, *bleaching_bin*, with three levels: Low (<1%), Medium (1–30%), and High (>30%). The new class distribution, while still somewhat imbalanced, would help improve model learnability and ecological interpretability (Appx. Figure 1). Observations missing the target (6,837 rows; 17%) were removed, leaving 34,468 rows and 41 predictors. The dataset was then partitioned into a 60/20/20 split, as summarized in Table 1.

Exploratory analysis of the BCO-DMO dataset showed that low-severity reefs dominated, and thermal features exhibited nonlinear, monotonic relationships with bleaching severity, making Spearman correlation the appropriate method for our correlation matrix (Appx. Figure 2). Most features showed only weak correlations with bleaching (0.10–0.39), with moderate associations appearing mainly in thermal-stress variables such as ssta_dhwmean, ssta_frequencymean, and temperature_maximum. Spatial variables showed poor correlation to bleaching severity, while environmental factors like turbidity and windspeed displayed broader variation across classes. Feature family analysis revealed redundancy within the SSTA and TSA groups, suggesting that our models should retain only the strongest representatives (e.g., ssta_dhwmean, ssta_maximum, climsst) while dropping highly correlated alternatives, whereas tree-based models can safely keep all features due to robustness to multicollinearity. Distribution plots (Appx. Figure 3) confirmed substantial overlap across classes for most predictors, indicating that bleaching arises from complex, nonlinear interactions rather than clear linear separability.

The Healthy and Bleached Corals Image Classification dataset from Kaggle (2), contains 923 labeled images of healthy (n = 438) and bleached (n = 485) corals sourced via the Flickr API. All images were resized to 224×224×3 pixels and loaded into a standardized array of shape (923, 224, 224, 3), with a corresponding label vector of length 923. The dataset was partitioned into a 60/20/20 split, as summarized in Table 1. The class distribution (Appx. Figure 4) shows mild imbalance (52.5 percent bleached vs. 47.5 percent healthy), which is unlikely to substantially bias model predictions. As part of the EDA, we inspected image shapes to verify consistent dimensions, and visually examined the first few images from each class to confirm that resizing preserved image quality and that labels accurately reflected the visual distinction between healthy (typically brown/green) and bleached (paler/white) corals. We also examined pixel-intensity distributions across color channels (Appx. Figure 5, Figure 6) and found that mean pixel values were only slightly higher for healthy corals (approximately 98.4) than for bleached corals (96.6), indicating subtle differences in overall color composition between classes. The main challenge for the image dataset involved initial loading errors due to non-uniform resolutions, which was resolved by resizing all images to a common resolution during ingestion; this same step ensured compatibility with downstream convolutional models and provided a clean basis for subsequent augmentation and training.

Data Subset	BCO Environmental - Size	Coral Images - Size
Training	(20680, 41)	(553, 224, 224, 3)
Validation	(6893, 41)	(185, 224, 224, 3)
Test	(6895, 41)	(185, 224, 224, 3)

Table 1: Data subset sizes

Methods

To predict coral bleaching severity from structured reef-site data, we evaluated three models: multiclass logistic regression, a feed-forward neural network (FFNN), and a histogram gradient boosting classifier (HGBT). The target variable consisted of three bleaching severity classes (0= low, 1= medium, and 2= high), derived from percent bleaching observations.

Multiclass logistic regression was used as the baseline model because it provides a simple and interpretable benchmark for classification. Because linear models are sensitive to multicollinearity and feature magnitude, we performed model-specific feature selection. Using Spearman's rank correlation we retained features with meaningful monotonic relationships to bleaching severity. Variance inflation factor (VIF) analysis then identified and removed multicollinear predictors, yielding 11 numeric features and 2 categorical features. The remaining features were preprocessed using a scikit-learn Pipeline with median imputation, StandardScaler and one-hot encoding.

We next implemented a feed-forward neural network to better capture nonlinear relationships among environmental predictors. The network consists of fully connected layers that transform the input data through learned hidden representations before producing class probabilities at the output layer. Neural networks are well suited for problems where predictor interactions are complex and relationships are not well approximated by linear decision boundaries. Before building the network, we used SHAP (SHapley Additive exPlanations) values to assess feature importance (Appx. Figure 7), which guided reduction to the top 20 predictive features. Boolean categorical columns were converted to float values, and missing values were mean-imputed. Continuous features were standardized using StandardScaler to prevent gradient issues during backpropagation. The initial network architecture consisted of an input layer and four hidden layers with 128, 64, and 32 neurons respectively, each using ReLU activation, while the output layer used softmax activation to produce class probabilities.

Our final model was a histogram gradient boosting classifier. This tree-based ensemble method builds shallow decision trees sequentially, with each new tree correcting errors made by the prior ensemble. Histogram-based binning improves computational efficiency while retaining predictive power. Gradient boosting is especially effective for structured tabular datasets as it can model nonlinear effects, feature interactions, and threshold behavior without requiring extensive manual feature engineering. Hyperparameter tuning was performed via parameter grid over learning_rate, max_iter, min_samples_leaf, max_leaf_nodes, and max_depth. All HGBT models used log_loss and balanced class weights.

To classify coral images as healthy or bleached, we evaluated three image-based classification models: binary logistic regression, a convolutional neural network (CNN), and a tuned CNN. All images were resized to a common resolution of 224×224×3 and normalized prior to training. Model performance was evaluated using classification accuracy on validation and test datasets. Binary logistic regression was used as the baseline model. For this approach, each image was flattened into a one-dimensional vector of pixel values and used to estimate the probability that the image belonged to the bleached class. The training dataset was also expanded with images augmented through brightness and contrast adjustment and randomly flipped left to right. We implemented this initial model in TensorFlow using a single dense layer with one unit, sigmoid activation, and bias. The model was compiled with SGD optimization and binary cross-entropy loss. Although simple and computationally efficient, this method does not preserve spatial relationships between neighboring pixels.

We then implemented convolutional neural networks, which are optimal for image data. CNNs use convolutional filters to detect local visual patterns such as edges, textures, and shapes, while pooling layers reduce dimensionality and improve robustness. These learned features are then passed to dense layers for final classification. To further improve performance, we conducted hyperparameter tuning on the CNN architecture. Tuned parameters included dense layer size, dropout rate, learning rate, batch size, and number of training epochs. Our initial architecture consisted of three convolutional blocks of 32, 64 and 128 filters and ReLU activation. Dropout was used as a regularization strategy to reduce overfitting and improve generalization to unseen images. The output layer used a single unit with sigmoid activation for binary classification.

Experiments, Results, and Discussion

For the environmental bleaching severity task, we compared baseline, intermediate, and final models using the same train, validation, and test partitions to ensure fair evaluation. The goal of these experiments was to determine whether increasingly flexible models could better capture the ecological complexity underlying bleaching outcomes.

Multiclass Logistic Regression (Baseline): Three experiments were conducted: baseline configuration with median imputation, L2 regularization(c=1.0) and balanced class weights; (2) stronger regularization (c=0.1); and mean imputation. All experiments used the reduced feature set derived from Spearman Correlation and VIF analysis yielding similar performance: 66% accuracy with identical failure patterns. The medium bleaching class had very low recall (0.39), showing that the model misses most moderate bleaching observations. The F1 score remained 58% across experiments confirming that linear decision boundaries are insufficient for this task.

Feed-Forward Neural Network: For this model, we aimed to test whether a nonlinear architecture could improve predictive performance. Multiple configurations were explored through ablation experiments, allowing comparison of architectures and hyperparameters based on training versus validation metrics. The ablation study (Appx. Figure 8) included a variety of hidden layer sizes and learning rates, a comparison of SGD and Adam optimizers, rectified linear unit (ReLu) versus hyperbolic tangent (tanh) activation functions, and various numbers of training epochs. Experimentation with class weighting was also attempted due to class imbalance, though this strategy caused a considerable drop in performance metrics and was ultimately abandoned. Dimensionality reduction ultimately proved key to optimizing the FFNN performance, as the initial iteration of this model utilized all 41 features of the dataset. With the selection of the top 20 features, identified via SHAP feature importance analysis, immediate improvements in validation accuracy were noted. The ablation study process was then repeated and fine-tuned, with the final model selection consisting of Adam optimizer, hyperbolic tangent activation, a learning rate of 0.00075, and six hidden layers of 1024, 512, 256, 128, 62, & 32 nodes respectively. Training with this model configuration for 16 epochs resulted in the optimal balance of training versus validation metrics without overfitting.

Histogram Gradient Boosting: We applied a structured hyperparameter search on the validation set, evaluating each configuration by its macro F1 score and the training–validation accuracy gap as an indicator of overfitting. While the GB baseline and first tuned model achieved slightly higher macro F1 scores (>0.81), both showed large accuracy between sets gaps of about 12%, suggesting poor generalization. The last tuned model (GB_tuned2) reduced this gap to 6.9% while still maintaining a solid macro F1 score of 0.757. This improvement comes from its more constrained architecture: learning rate set to 0.03, increased minimum samples required per leaf to 10, cap of 20 leaf nodes per tree, and extended training to 300 boosting iterations (Appx. Table 4). Together, these changes produce shallower, more regularized trees that learn more gradually and avoid complex splits seen in the earlier models. Given the priority of model generalization, GB_tuned2 was chosen as the final model for evaluation due to minimized overfitting.

Final models evaluation: Because the bleaching classes were imbalanced, evaluation emphasized F1-macro score in addition to accuracy. Accuracy measures overall correctness, while F1-macro score provides a stronger assessment of precision and recall across classes. This was particularly important because correctly identifying medium and high-risk bleaching events is more meaningful than only predicting the majority low-risk class. Although the FFNN model achieved a slightly higher macro-F1 score than the histogram gradient boosting model as shown in Table 2, the boosting model delivered stronger overall test accuracy (79.75%) while still maintaining balance across classes. This combination provides a better general predictive performance without disproportionately sacrificing minority-class behavior which made the gradient boosting model the more reliable choice as our final model.

Model	Training Accuracy	Validation Accuracy	Test Accuracy	F1-Macro (Test Data)
Multiclass Logistic Regression	65.96%	66.45%	66.55%	58.08%
Feed Forward Neural Network	78.65%	76.93%	76.49%	78.00%
Hist. Gradient Boosting Classifier	85.77%	79.85%	79.75%	75.68%

Table 2: Environmental Dataset Results

For the image classification task between healthy and bleached coral, we evaluated three models: (1) **A binary logistic regression classifier, (2) convolutional neural network (CNN), and (3) a hyperparameter tuned CNN**. For the first model, we used flattened pixel values. This experiment established a simple baseline and tested whether raw pixel intensities alone were sufficient for accurate classification. For the improved model (CNN), we used resized coral images. This experiment evaluated whether learned spatial features could improve classification accuracy relative to the baseline model. We then performed hyperparameter tuning on the CNN by varying architectural and optimization settings, including dense layer size, dropout rate, learning rate, batch size, and number of epochs. Candidate models were compared using validation accuracy, and the best-performing configuration was selected for final test evaluation. The top-performing configuration used 256 dense units, dropout = 0.5, learning rate = 0.0005, batch size = 32, and 10 epochs (Appx. Table 6). The logistic regression model achieved 66.49% test accuracy, while the initial CNN substantially improved performance to 70.81% test accuracy. The tuned CNN improved training and validation results, indicating that the adjustment of our

hyperparameters during experimentation led to more refined results. Overall, these experiments demonstrate that preserving spatial image structure is critical for coral bleaching image classification and that convolutional models are far better suited to this task than the linear baseline. Evaluation metrics of all models are shown below in Table 3.

Model	Training Accuracy	Validation Accuracy	Test Accuracy
Logistic Regression	68.35%	70.81%	66.49%
CNN	71.52%	69.73%	70.81%
Tuned CNN	79.57%	74.05%	75.15%

Table 3: Image Dataset Results

Conclusion

This study showed that the ensemble method Histogram Gradient Boosting model outperformed all other approaches for the structured environmental dataset when predicting the three-class bleaching levels in the environmental dataset. For the coral image dataset, framed as a binary healthy or bleached output, the tuned CNN achieved the strongest performance relative to the logistic regression baseline. These results are achievable because of the following mechanisms: the HGBT model is well suited to noisy, nonlinear ecological relationships, and with a more constrained configuration, this allows its trees to learn more efficiently. Conversely, the CNN can extract spatial and color-based features from images, enabling gradual learning of color, texture, and spatial cues while limiting overfitting. This is particularly effective for coral reefs given their rich color patterns and complex structures and textures. Together, these properties allow both models to substantially outperform the logistic regression baselines and achieve greater generalizability.

This project faced several limitations that could be addressed with additional time or computational resources. The structured dataset exhibited significant class imbalance, which we addressed by using macro F1 scores for evaluation metrics, but more sophisticated techniques such as synthetic oversampling (SMOTE) or cost-sensitive learning may further improve the moderate-severity class recall. The BCO-DMO dataset also contained a large number of unlabeled rows which could be leveraged by semi-supervised or unsupervised approaches to learn richer feature representations. The image dataset was also relatively small, limiting the depth of CNNs we could train without aggressive dropout and augmentation. Access to larger labeled coral image collections would likely improve robustness and generalization to novel reef environments. Additionally, treating image and environmental data separately limits model capability, and a multimodal approach could better capture complementary signals. Future work should also explore advanced imbalance handling techniques and uncertainty estimation for real-world deployment. Despite these limitations, our results demonstrate that gradient boosting for environmental data and convolutional neural networks for imagery can effectively support early warning systems, targeted reef survey prioritization, and coral conservation planning.

Contributions

Marisa:

- **Notebooks:** coral_images_preprocessing.ipynb, coral_images_baseline.ipynb, BCO_FFNN_model.ipynb
- **Report:** Feed-forward neural network portion of Experiments section, editing and additions to remaining sections

Priyanka

- **Notebooks**: BCO_Prep_EDA.ipynb, BCO_baseline_final_model.ipynb, coral_images_EDA.ipynb
- **Report**: contributed to Abstract, Introduction, related work, and initial drafts of dataset, methods, experiments and conclusion.

Mackenzie

- **Notebooks**: BCO_Preprocessing.ipynb and coral_images_improvement_model.ipynb
- **Report**: contributed to methods, experiments, results, discussion, conclusion, and edited other sections

Paola

- **Notebooks**: BCO_EDA.ipynb, BCO_Final_Preprocessing_EDA.ipynb and BCO_HGB_model
- **Report**: Contributed to dataset, experiments, results, conclusion and edited other sections

Github Repository: https://github.com/mackenzie-henderson/DATA-207-Final-Project

References:

1. Cheung, M. W. M., Chaloupka, M., Hock, K., & Mumby, P. J. (2025). *Moving beyond temperature metrics in coral bleaching prediction using interpretable machine learning*. Global Ecology and Biogeography, 34(8), e70105. https://www.researchgate.net/publication/394532865
2. Deep Learning Models for Coral Bleaching Classification in Multi-Condition Underwater Image Datasets: Julio Jerison E. Macrohon, Gordon Hung https://arxiv.org/abs/2511.00021
3. Reef Check, Donner et al. (2017), McClanahan et al. (2019), AGRRA (https://www.agrra.org)
4. FRRP (https://ocean.floridamarine.org/FRRP/Home/Reports), Safaie et al. (2018), and Kumagai et al. (2018).

Data Sources:

1. BCO-DMO Global Bleaching & Environmental Dataset
2. Healthy and Bleached Corals Image Classification – Kaggle Dataset

Appendix:

Figure 1: Class Distribution - Environmental Dataset

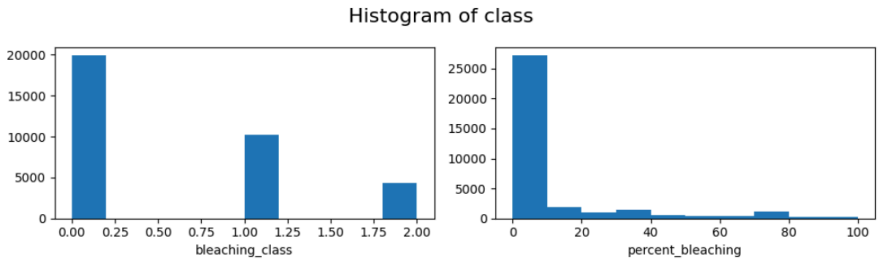


Figure 2: Correlation Matrices - Environmental Dataset

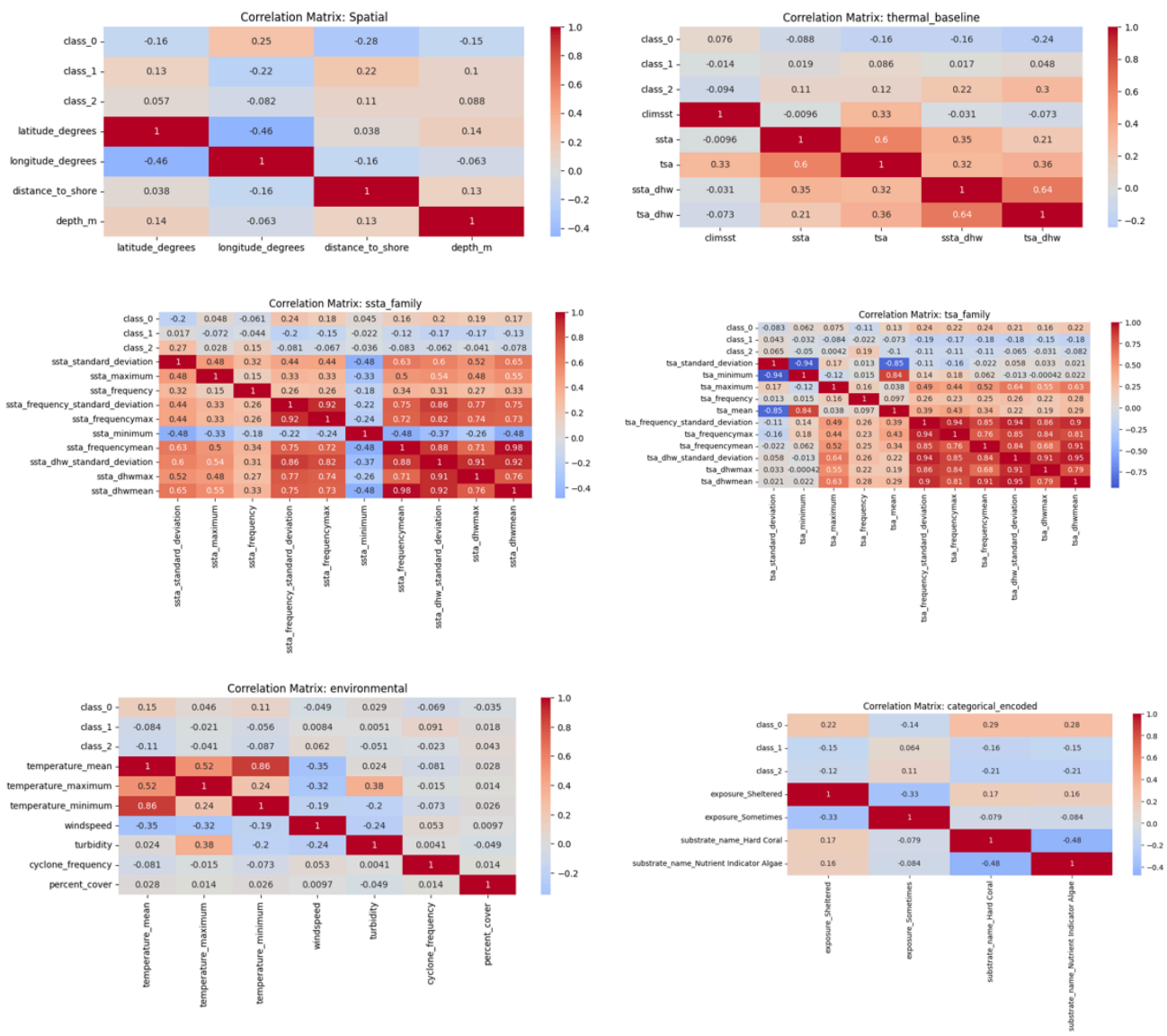
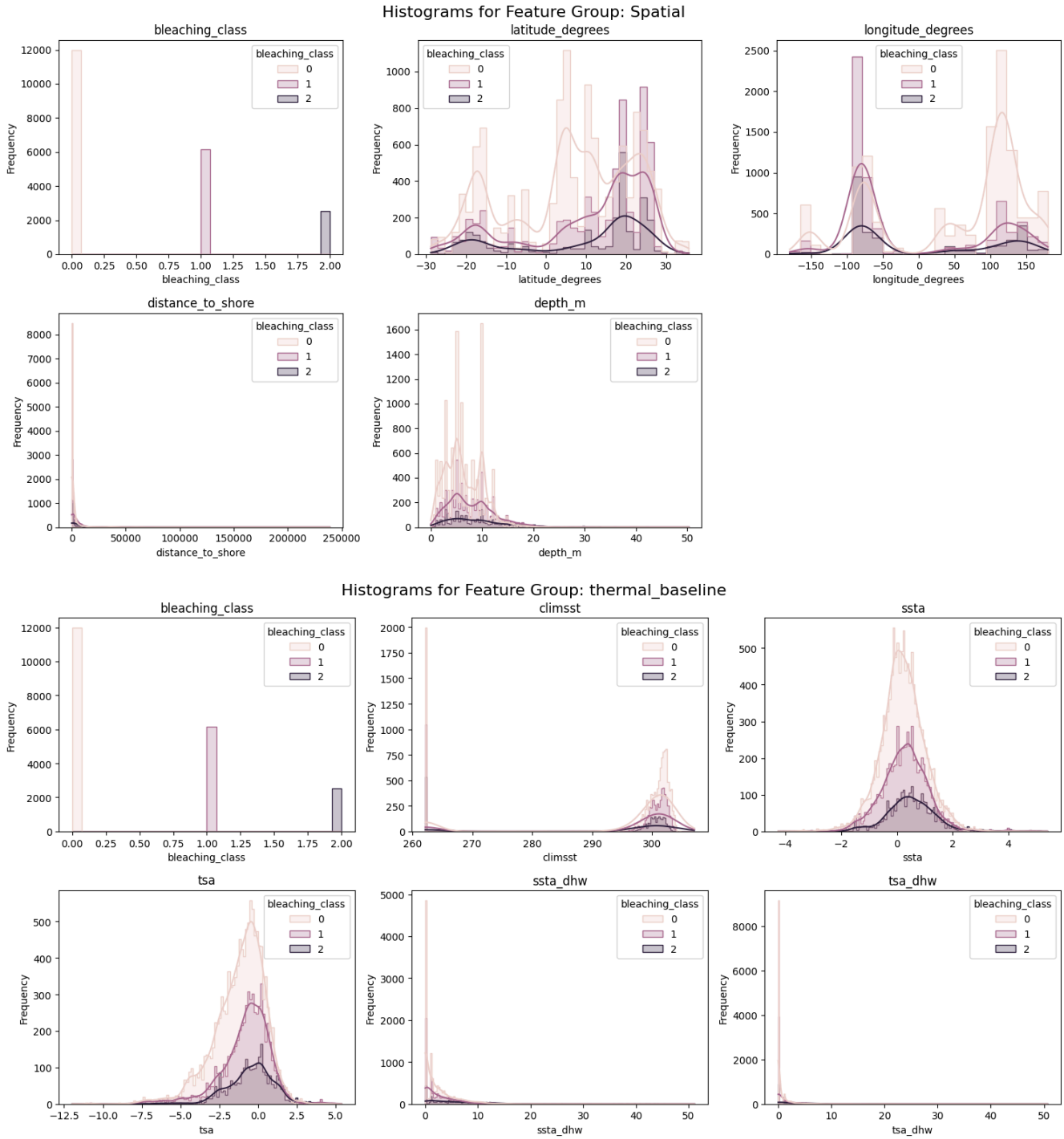
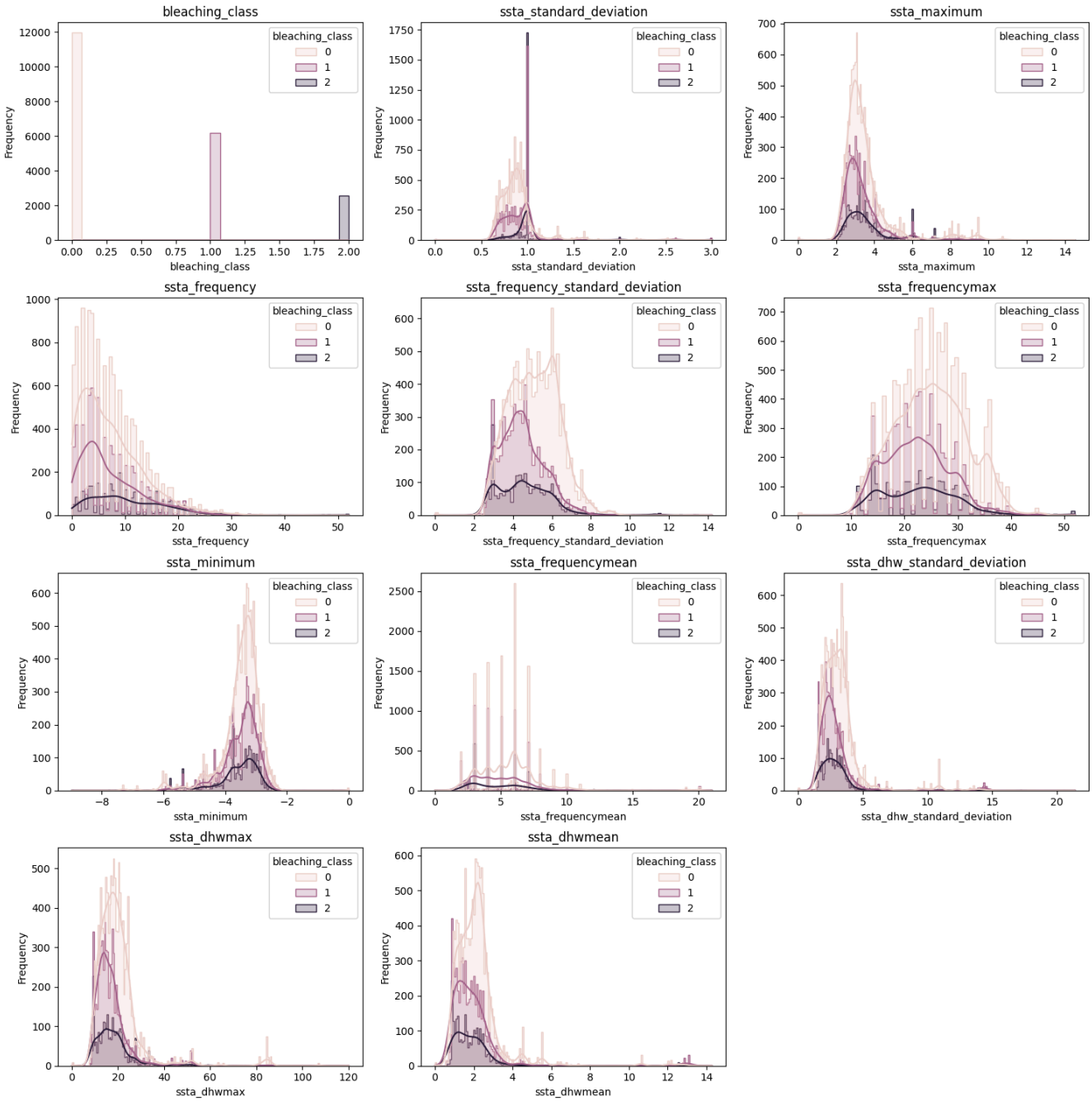


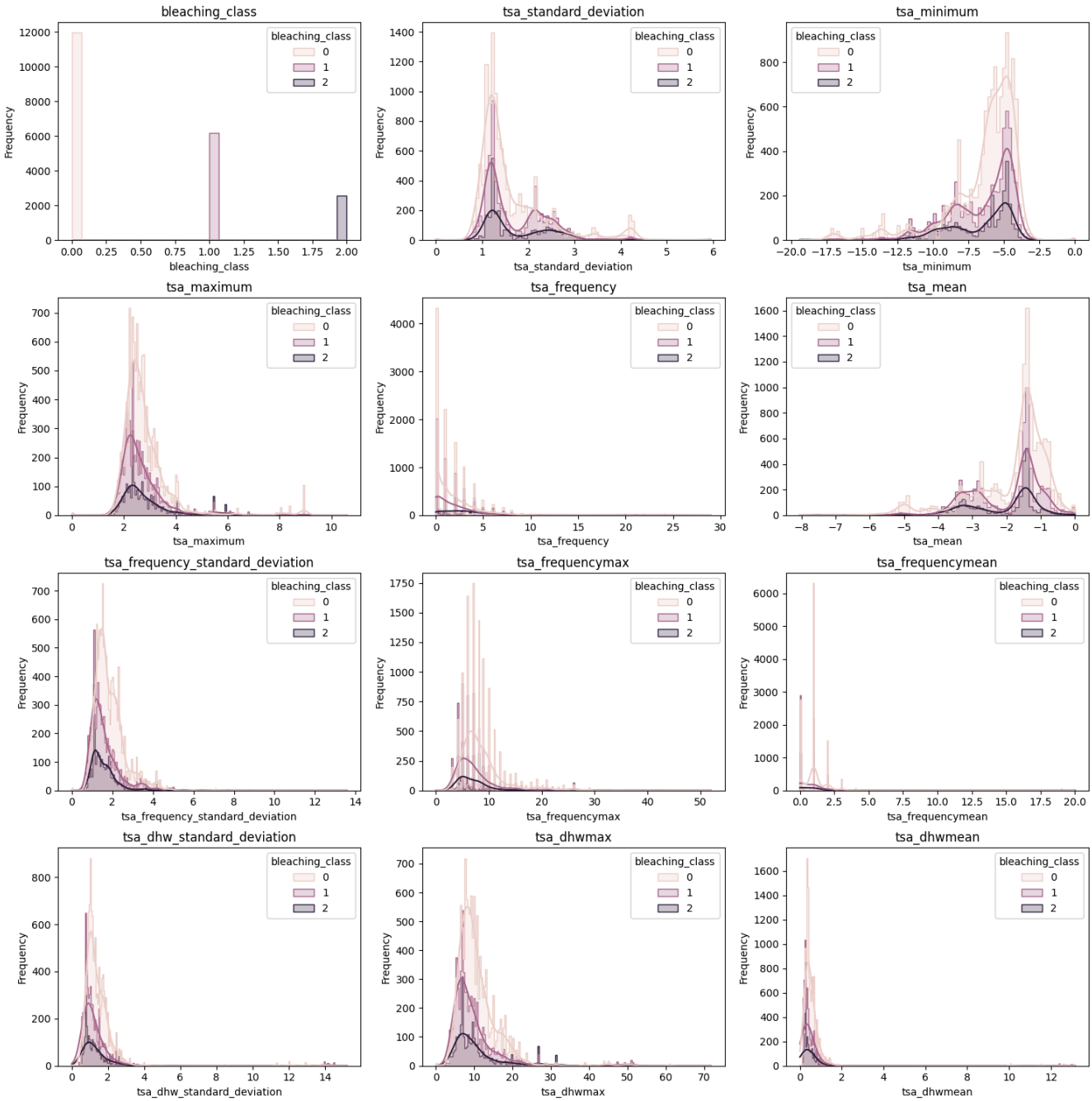
Figure 3: Histogram for feature groups - Environmental Dataset



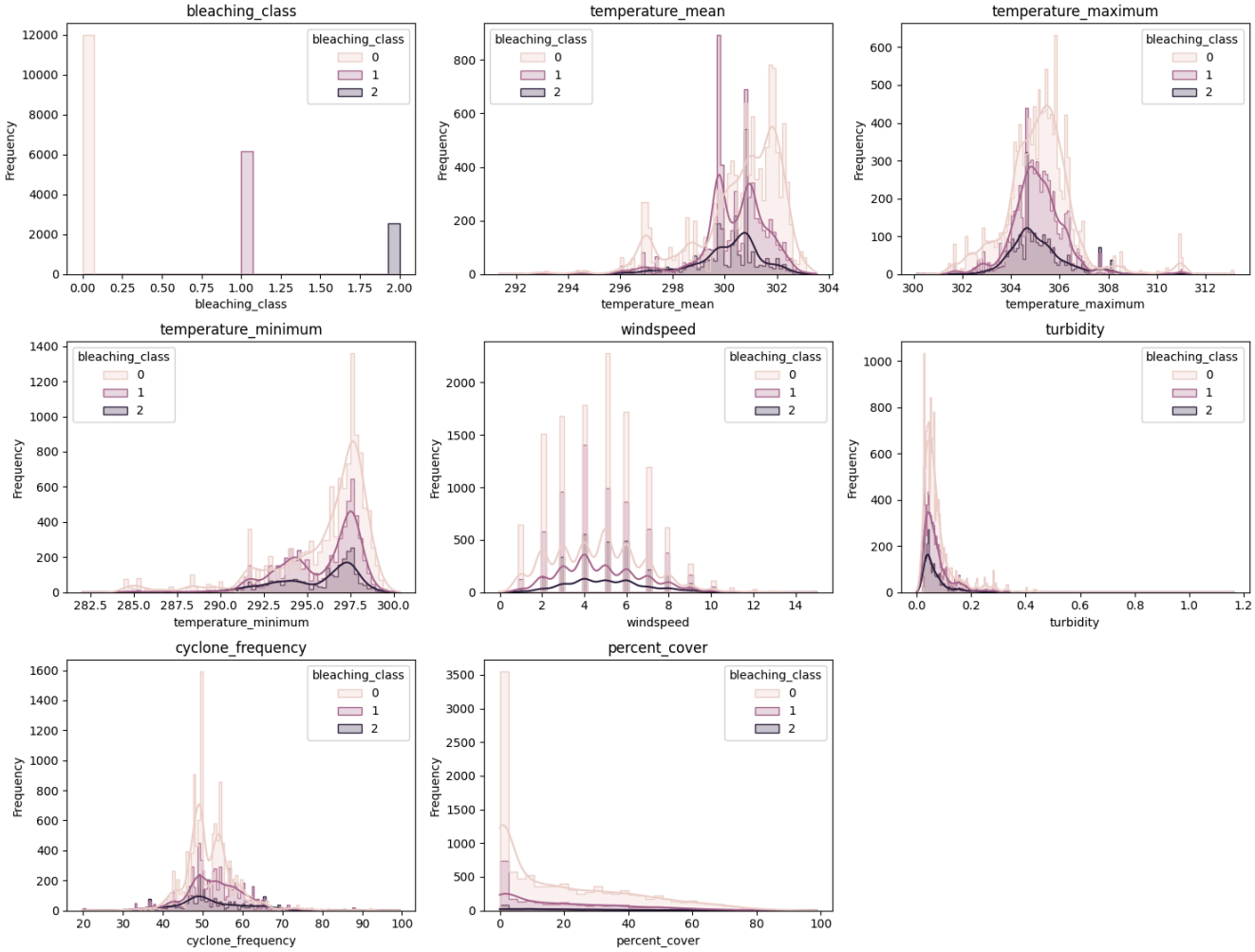
Histograms for Feature Group: ssta_family



Histograms for Feature Group: tsa_family



Histograms for Feature Group: environmental



Histograms for Feature Group: categorical_encoded

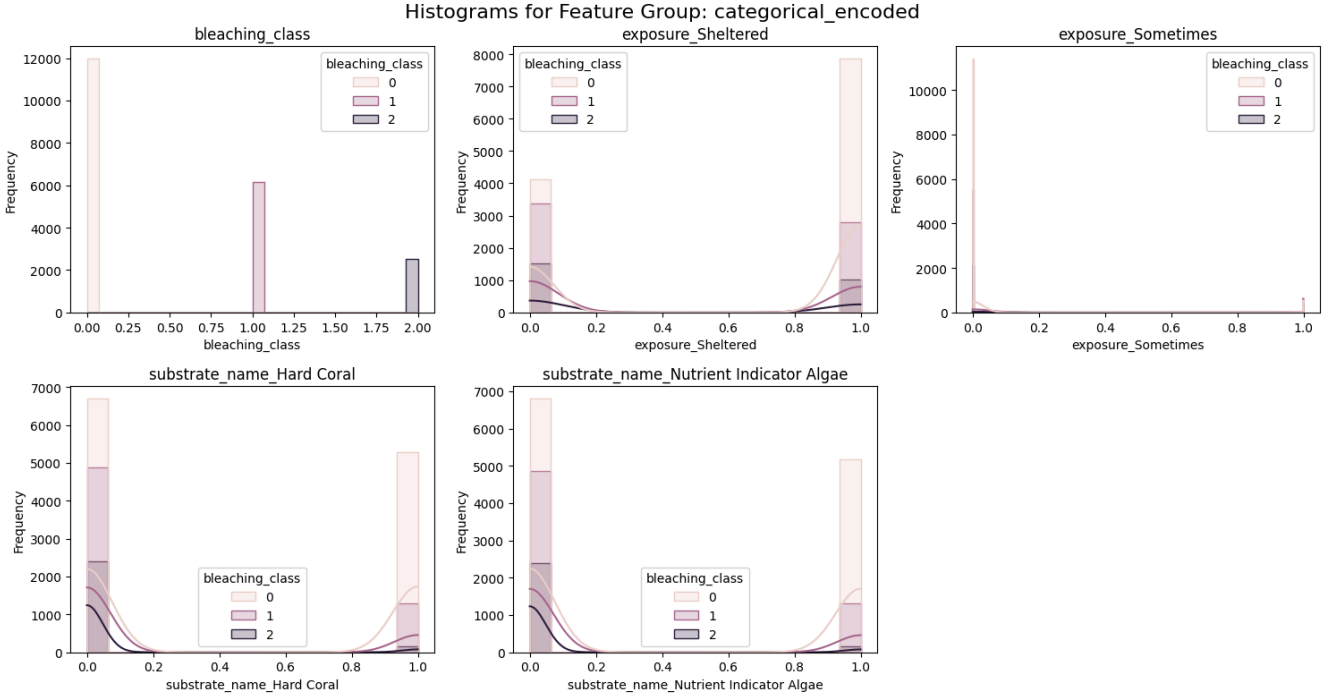


Figure 4: Class Distribution - Coral Images Dataset

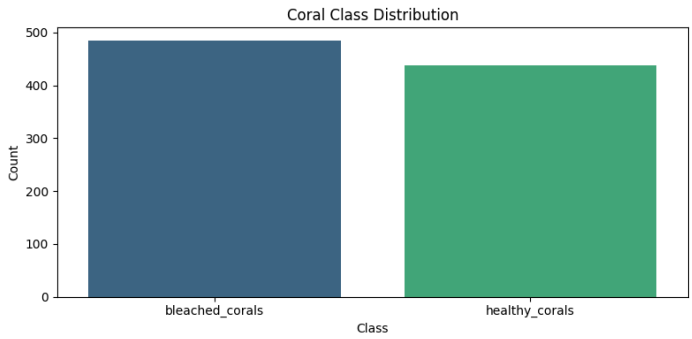


Figure 5: Mean Pixel Values - Healthy Coral Images

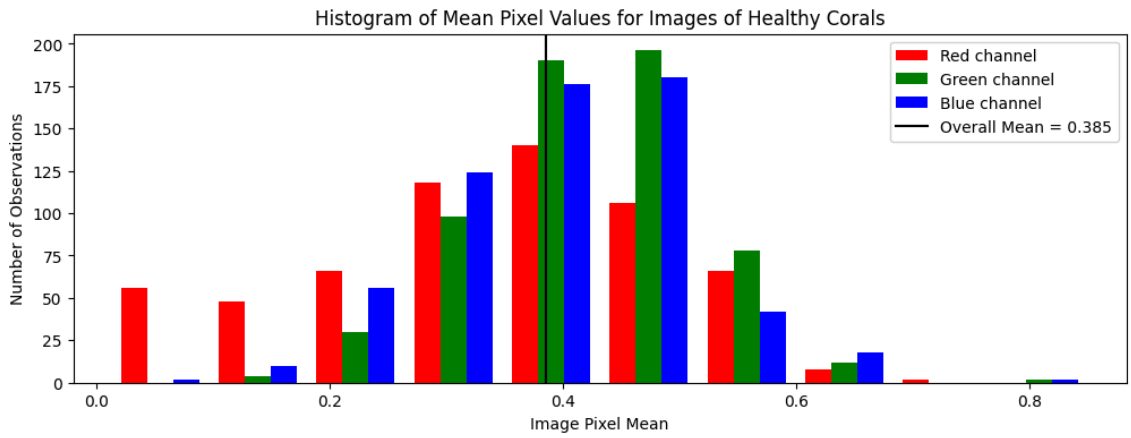


Figure 6: Mean Pixel Values - Bleached Coral Images

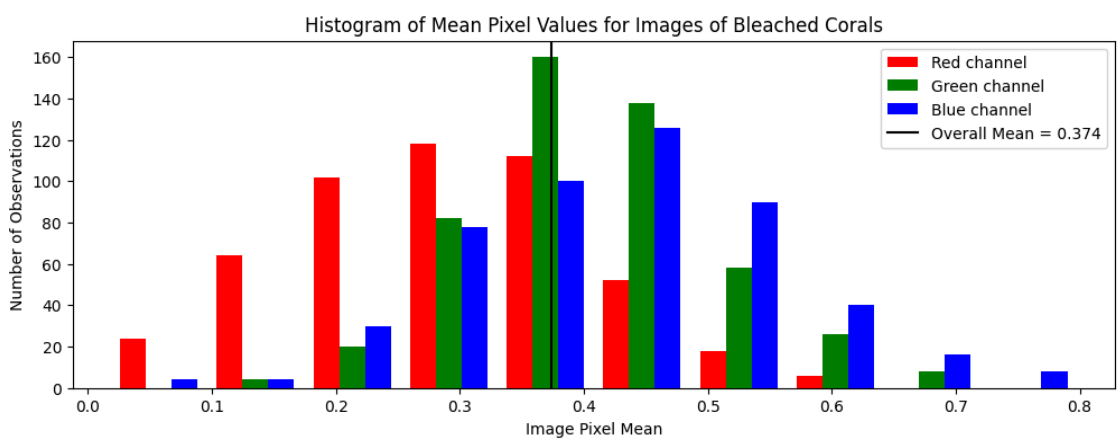


Figure 7: SHAP summary per class

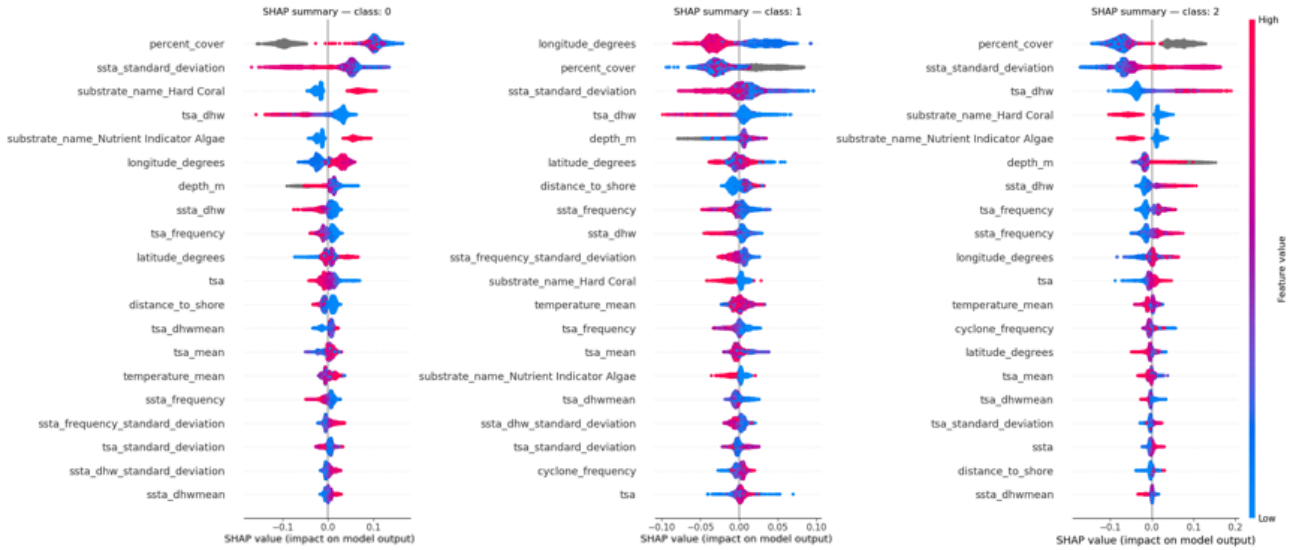


Figure 8. Ablation study for Feed-Forward Neural Network Model (BCO Environmental Dataset)

Model	Hidden layer sizes	Activation	Optimizer	Learning rate	Epochs	Training accuracy	Validation accuracy
model_tf1	[128,62,32]	relu	Adam	0.0001	15	0.6867	0.6775
model_tf2	[256,128,62,32]	relu	Adam	0.0001	15	0.4965	0.4362
model_tf3	[512,256,128,62,32]	relu	Adam	0.001	15	0.5246	0.6494
model_tf4	[62,32]	tanh	Adam	0.001	15	0.7600	0.7529
model_tf5	[62,32]	tanh	Adam	0.001	30	0.7827	0.7585
model_tf6	[128,62,32]	tanh	Adam	0.001	30	0.8248	0.7720
model_tf7	[256,128,62,32]	tanh	Adam	0.001	25	0.8596	0.7594
model_tf8	[128,62,32]	tanh	Adam	0.005	20	0.7926	0.7628
model_tf9	[128,62,32]	tanh	Adam	0.00005	40	0.7490	0.7418
model_tf10	[128,62,32]	tanh	Adam	0.000075	40	0.7521	0.7439
model_tf11	[512,256,128,62,32]	tanh	Adam	0.001	10	0.7988	0.7577
model_tf12	[1024,512,256,128,62,32]	tanh	Adam	0.001	15	0.8014	0.7660
model_tf13	[1024,512,256,128,62,32]	tanh	Adam	0.0001	30	0.8457	0.7822
model_tf14	[2048,1024,512,256,128,62,32]	tanh	Adam	0.000075	12	0.7861	0.7691
model_tf15	[62,32]	tanh	Adam	0.001	50	0.8010	0.7662
model_tf16	[1024,512,256,128,62,32]	tanh	Adam	0.0001	16	0.7865	0.7693

Figure 9. Test Data Confusion Matrix for Feed-Forward Neural Network Model (BCO Environmental Dataset)

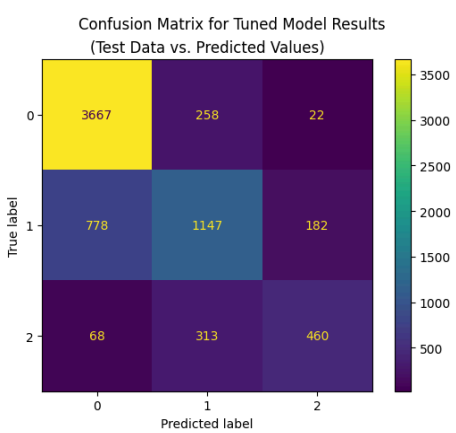


Figure 10: Test Data Confusion Matrix for CNN Model and Tuned CNN Model (Image Dataset)

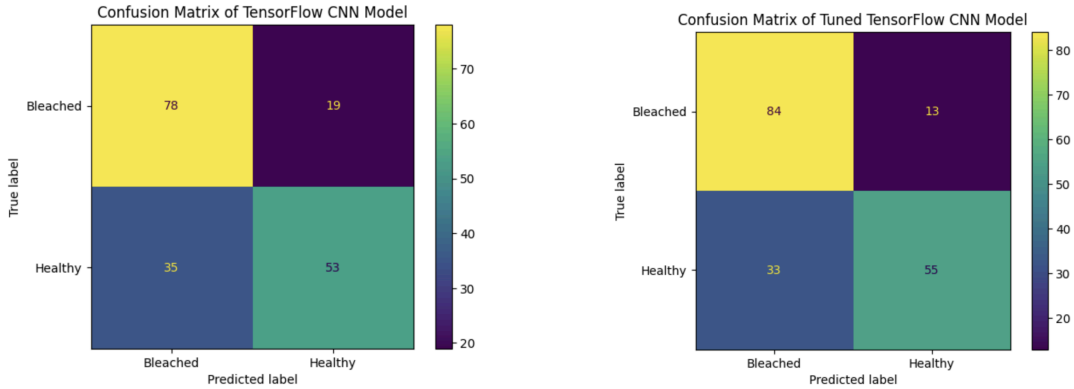


Table 4. Hyperparameter Tuning for the Histogram Gradient Boosting Model

Tuned Model	Training accuracy	Validation accuracy	Gap_%	Precision_ macro	Recall_ macro	F1_macro	Precision_per_ class_[0,1,2]	Recall_per_class _[0,1,2]	F1_per_class_[0, 1,2]
GB_baseline	0.9807	0.8599	12.32	0.8148	0.8228	0.8186	[0.9289, 0.7777, 0.7378]	[0.9268, 0.7609, 0.7808]	[0.9279, 0.7692, 0.7587]
GB_tuned	0.9802	0.8599	12.27	0.8149	0.8246	0.8193	[0.9273, 0.7814, 0.736]	[0.9276, 0.7544, 0.7919]	[0.9275, 0.7677, 0.7629]
GB_tuned2	0.8577	0.7985	6.9	0.7438	0.7764	0.7536	[0.881, 0.7096, 0.641]	[0.8909, 0.5915, 0.8468]	[0.8859, 0.6452, 0.7296]

Table 5. Final model evaluation for the Histogram Gradient Boosting Model

Data Set	Accuracy	Precision _macro	Recall_ macro	F1_macro	Precision_per_ class_[0,1,2]	Recall_per_class_ [0,1,2]	F1_per_class_[0,1,2]	Support
Training Data	0.8577	0.817	0.8558	0.8301	[0.914, 0.816, 0.721]	[0.9172, 0.7068, 0.9434]	[0.9156, 0.7575, 0.8173]	[11975, 6163, 2542]
Validation Data	0.7985	0.7438	0.7764	0.7536	[0.881, 0.7096, 0.641]	[0.8909, 0.5915, 0.8468]	[0.8859, 0.6452, 0.7296]	[4004, 1995, 894]
Test Data	0.7975	0.7456	0.7835	0.7568	[0.8737, 0.7289, 0.6342]	[0.8885, 0.6023, 0.8597]	[0.881, 0.6596, 0.7299]	[3947, 2107, 841]

Table 6. Hyperparameter Tuning for the Tuned CNN Model

	dense_units	dropout_rate	learning_rate	batch_size	epochs	train_accuracy	val_accuracy
0	128	0.3	0.0005	32	10	0.795660	0.756757
1	64	0.3	0.0010	32	8	0.740506	0.729730
2	128	0.5	0.0010	32	10	0.744123	0.729730
3	256	0.5	0.0005	32	10	0.760398	0.724324